

HeroPips — Product Requirements Document (v1.0)

Product: HeroPips (heropips.com) — AI-assisted trading automation, academy & signal platform **Owner:** Founder / CEO **Status:** Draft for stakeholder review **Date:** July 24, 2026 **Beta window:** October – December 2026 · **GA target:** January 2027

Version	Date	Author	Change
0.1	Jul 24, 2026	Founder + Product	Initial end-to-end draft from strategy research
1.0	—	—	Approved baseline (pending review)

Related documents: HeroPips Investor-Grade Productization & Business Plan (strategy source); Technical Architecture Spec (to be derived from §12); Legal/Compliance Memo (to be commissioned per §16).

Branding decision (locked): The product ships as **HeroPips** on **heropips.com**. hellopips.com is registered defensively and 301-redirects to heropips.com. No dual branding.

1. Product Overview

HeroPips takes a complete beginner from **zero to hero**: learn trading in a structured academy, practice risk-free with paper trading, then connect a real broker or exchange (MetaTrader 5, cTrader, or crypto exchanges such as Binance) and trade with automation, AI-generated signals, TradingView alert execution, and a disciplined risk-automation layer that enforces the rules most retail traders break.

HeroPips is a **self-directed software tool**. It does not provide investment advice, does not manage or custody client funds, and does not guarantee results. Users connect their own broker accounts, define their own rules, and control every trading decision. This positioning is a hard product constraint, not marketing copy (see §16).

Positioning statement: For aspiring and intermediate retail traders in MENA, India, and SEA who are underserved by expert-only automation tools and predatory signal sellers, HeroPips is the education-plus-automation platform that combines a beginner academy, paper trading, broker-agnostic execution, and prop-firm-grade risk guards in one subscription — unlike TradersPost (execution only), 3Commas/Cryptohopper (crypto only), or copy-trading platforms (no education, no user control).

Revenue streams (in scope for this PRD): SaaS subscriptions (Free/Starter/Pro/Elite), Academy sales, capped lifetime deal (beta only), IB/broker rebate revenue, affiliate-driven growth. Phase 3 adds white-label licensing and a signal-marketplace take-rate.

2. Problem Statement

Three connected problems, one platform:

1. **Beginners have no honest bridge from zero to live trading.** Education is either free-but-shallow or high-ticket and predatory (the FTC's \$362M judgment against Online Trading Academy is the cautionary benchmark). Nothing takes a novice from lessons → paper practice → carefully guarded live trading in one continuous journey.
2. **Strategy owners cannot execute automatically.** Millions of TradingView users receive alerts they must trade by hand — slow, error-prone, emotional. Existing bridges (TradersPost, Alertatron) are crypto/US-centric and ignore the MT5 forex world where most MENA/India/SEA retail volume lives.
3. **Retail and prop-firm traders lose to indiscipline, not analysis.** Blown daily-loss limits, oversized positions, revenge trades, stale pending orders, news-spike slippage. Prop-firm pass rates average 5-10%, mostly due to rule violations. Nobody sells affordable, broker-agnostic "discipline as a service."

Cost of not solving: users churn to fragmented point tools; the education funnel (our cheapest acquisition channel) leaks; and the defensible wedge — the full zero-to-hero journey — is ceded to whoever combines these first.

3. Goals & Success Metrics

3.1 Goals (12-month horizon, GA + 12mo)

#	Goal	Type	Target
G1	Prove activation: users reach "first trade" fast	User	≥ 40% of signups place a first paper or live trade within 24h
G2	Prove the education→automation funnel	Business	≥ 3% free→paid conversion; ≥ 25% of paid users completed ≥ 3 academy lessons first
G3	Build durable recurring revenue	Business	\$40k MRR by mid-2027; monthly logo churn ≤ 6%
G4	Monetize free users via IB rebates	Business	≥ \$10k/month broker rebate revenue by Q2 2027
G5	Make risk automation the retention moat	User	≥ 60% of live-connected users have ≥ 1 active risk guard; guarded users churn ≥ 30% less
G6	Ship a viral loop that is legally clean	Business	Referral K-factor ≥ 0.3 during beta with a max 2-level, non-cash-default program

3.2 Metric definitions

- **Activation:** account created AND (paper account funded OR broker connected) AND first order placed, within 24h of signup.
- **Leading indicators:** signup→connect rate, webhook success rate, signal-enable rate, lesson completion, guard adoption, time-to-first-trade.

- **Lagging indicators:** MRR, free→paid %, D30/D90 retention, churn, IB rebate revenue, NPS (target ≥ 40), LTV:CAC (target ≥ 3 by GA+12mo).
- **Measurement:** PostHog product events (§15) + Stripe + internal ledger; evaluated weekly during beta, monthly after GA.

4. Non-Goals (v1 — explicit)

#	Non-goal	Rationale
NG1	10-level cash referral program	Pyramid-scheme legal/reputational risk. System supports configurable depth (§E13) but product policy caps at 2 levels without legal sign-off.
NG2	Withdrawable bonus/points wallet	Triggers money-transmission / stored-value licensing. Points are non-cash credits only (§E12). Cash flows only through the approved affiliate rail (§E14).
NG3	Investment advice or discretionary trading	Regulatory (robo-advisory licensing). AI outputs are educational analysis; execution requires explicit user-defined rules or confirmation.
NG4	Custody of user funds	Users' money stays at their broker/exchange. HeroPips never holds balances; API keys are trade/read scope only, never withdrawal scope.
NG5	Full backtesting engine in v1	Large build; Phase 2. v1 ships runtime signal backtest (Pro+) only.
NG6	Mobile native apps in v1	Responsive web first; apps are Phase 3.
NG7	White-label tenant onboarding in v1	Architecture is multi-tenant-ready from day 1 (tenant_id everywhere, config-driven branding), but no external tenants before Phase 3 and ≥ 3 committed leads.
NG8	MT4 support	MT5-first; MT4 only if demanded post-GA.
NG9	Profit claims / "guaranteed signals" marketing	Compliance and brand integrity. All performance shown with disclaimers and verified data only.

5. Personas

ID	Persona	Profile	Primary jobs-to-be-done	Key tier
P1	Aisha — the Zero	24, Dubai, salaried, no trading background; found HeroPips via a reel	Learn safely; practice without losing money; feel progress (points, levels)	Free → Starter
P2	Rahul — MT5 forex trader	31, Mumbai; 2 yrs on MT5; loses to indiscipline	Auto risk sizing, drawdown stop, news filter, journal; connect existing broker in minutes	Starter → Pro
P3	Marco — TradingView strategist	38, remote; Pine Script strategies; hand-trades alerts	Fire alerts → real orders on MT5 + Binance; idempotent, fast, with SL/TP attached	Pro
P4	Chen — crypto trader	27, KL; Binance/Bybit; wants bots without 3Commas complexity	Connect via API keys, run predefined signals, panic-close, track PnL	Starter → Pro
P5	Omar — prop-firm challenger	29, Cairo; failed 2 FTMO challenges on daily-loss breaches	Hard rule guards that block violating orders; challenge-preset templates	Elite
P6	Sana — affiliate / IB	Runs a 40k-follower Telegram forex channel	Custom deal, sub-affiliates, real-time stats, reliable payouts	Affiliate portal
P7	Vik — white-label buyer (Phase 3)	Runs a trading academy; wants his own branded platform	Turnkey branded tenant, own pricing, own affiliates	White-label
P8	Admin/Ops (internal)	HeroPips staff	Configure everything (plans, signals, tenants, deals, compliance flags) without deploys	Admin console

5.1 Top user stories (priority order)

1. As **Aisha**, I want a guided path (lessons → quiz → paper trade) so that I can go from knowing nothing to placing my first practice trade in one sitting.
2. As **Rahul**, I want to connect my MT5 account with login/server credentials and see my live balance, positions, and history inside HeroPips within minutes, so that I trust the platform before automating anything.
3. As **Rahul**, I want every order auto-sized from a % risk I set, with a daily-loss circuit breaker, so that one bad day can't blow my account.

4. As **Marco**, I want a unique webhook URL + secret so that my TradingView alerts open/close positions on my broker within seconds, exactly once, even if TradingView retries.
 5. As **Chen**, I want to enable a predefined signal on my Binance account in confirm-mode first, so that I can watch it before letting it trade fully automatically.
 6. As **Omar**, I want an "FTMO-style challenge guard" preset that blocks any order that would breach daily or max drawdown, so that I stop failing challenges on rule violations.
 7. As **Aisha**, I want to earn points for lessons, streaks, and paper milestones and redeem them for subscription time, so that progress feels rewarding without being a money game.
 8. As **Sana**, I want my own negotiated commission structure and links, with a live dashboard and a monthly payout I can trust, so that I promote HeroPips over competitors.
 9. As **Marco**, I want a runtime backtest of any signal config over historical data (Pro), so that I can sanity-check a setup before enabling it live.
 10. As **Admin**, I want plans, feature flags, signal library, referral depth, affiliate deals, and branding to be configurable in the admin console (backed by config JSON), so that we never hard-code business rules.
 11. As **Aisha**, I want a shareable results card of my paper-trading month so that I can show friends my progress (with automatic risk disclaimers).
 12. As **Rahul**, I want clear notification of every automated action (order placed/modified/rejected, guard triggered) by email and in-app, so that automation never feels like a black box.
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6. Release Plan & Scope Summary

Phase	Window	Theme	Ships
Phase 0 — Foundation	Aug - Sep 2026	Build core rails	Auth, billing skeleton, MT5 connect (MetaAPI), account mirroring, manual trade ticket, Academy v0 (10 lessons), paper trading v0, admin console v0, event instrumentation
Phase 1 — Closed Beta	Oct 2026	Prove activation	Signal engine (predefined + config v1), TradingView webhooks, risk guard suite v1, points v1, referral (2- level) v1, dashboards/journal v1; 500 invited users
Phase 1 — Open Beta	Nov - Dec 2026	Prove willingness to pay	Paid tiers live, capped Lifetime Deal ($\leq$ 500 seats), CCXT (Binance + Bybit), IB rebate onboarding, share cards v1, affiliate portal v1
GA v1.0	Jan 2027	Public launch	Hardening, runtime signal backtest (Pro+), cTrader connect (read + basic execution), Academy full curriculum, annual plans
Phase 2	2027 H1- H2	Depth	Full backtesting engine, signal marketplace (follow/publish, 20% take-rate), paper-trading tournaments, more exchanges, advanced guards, AR/HI localization
Phase 3	Q4 2027+	Scale	White-label tenants, mobile apps, master-IB tooling, self-hosted MT5 bridge (margin lever), enterprise API

Hard constraints: beta must end by Dec 31, 2026 (founder commitment); LTD is beta-only and permanently capped; live-broker connect is never available on the Free tier (unit-economics gate — MetaAPI cost is per connected account).

7. Packaging, Pricing & Entitlements

7.1 Tiers (monthly, USD; annual = 20% discount)

Capability	Free	Starter \$29	Pro \$79	Elite \$199
Academy core lessons + quizzes	✓	✓	✓	✓
Full Academy curriculum	—	✓	✓	✓
Paper trading	✓ (1 account)	✓	✓ (3 accounts)	✓ (5 accounts)
Live broker/exchange connections	0 (demo/paper only)	1	3	10
Predefined signals	View, 15-min delayed	✓ real-time, 3 active	✓ 10 active	✓ unlimited
Custom signal builder	—	1 config	✓ 10 configs	✓ unlimited
TradingView webhook automation	—	100 alerts/mo	1,000 alerts/mo	10,000 alerts/mo
Per-trade AI tips	—	✓	✓	✓
Risk guards — basic (SL/TP attach, panic close, basic trailing)	✓ (paper)	✓	✓	✓
Risk guards — premium (risk-% sizing, drawdown breaker, break-even, partial TP, ATR trailing, news & session filters, stale-order auto-cancel, slippage cap, max-concurrent)	—	Partial (sizing + drawdown)	✓ full	✓ full
Prop-firm challenge guard presets	—	—	—	✓
Runtime signal backtest	—	—	✓	✓
Signal lists: follow	view only	1	5	unlimited + publish (P2)
Priority execution queue & support	—	—	✓	✓
Seats / API access	—	—	—	Early API (read)

7.2 Other SKUs

- **Academy "Zero-to-Hero" one-time:** \$249 (range \$199–\$299, A/B in beta). Included free in annual Pro/Elite.
- **Lifetime Deal (beta only):** \$499–\$799, hard cap 500 seats, grants "Pro-Lifetime" with fair-use limits (Pro entitlements, 3 live connections). Never repeated after GA.
- **IB rebate track (revenue, not SKU):** opt-in flow to open/link broker accounts via HeroPips IB links; disclosed per §16.
- **White-label (Phase 3):** \$5,000 setup + \$2,000–\$3,000/month + rev share (pricing owned by Phase 3 PRD addendum).

7.3 Entitlement requirements

- **E7.1 (P0)** All tier limits above are enforced server-side via a central entitlement service keyed by `(plan_id)`; no client-side-only gating.
- **E7.2 (P0)** Plans, prices, limits, and feature flags are defined in admin-editable config (DB-backed JSON), hot-reloaded without deploy.
- **E7.3 (P0)** Downgrade rules: excess live connections are set read-only (not deleted); excess signal configs are paused; data is never destroyed by plan changes.
- **E7.4 (P1)** Grandfathering: LTD and beta pricing flags persist on the subscription record and survive plan-catalog changes.
- **E7.5 (P0)** Free tier can browse live-connect UI but hits a paywall at credential submission (conversion point, instrumented).

8. Functional Requirements by Epic

Requirement IDs: `(E<epic>.<n>)`; priority in parentheses (P0 = must-ship for the phase noted; P1 = fast-follow; P2 = future/architectural). Acceptance criteria (AC) listed for the highest-risk requirements.

E1 — Identity, Onboarding & Profiles (Phase 0)

Goal: friction-light signup that routes each persona to their fastest first win.

- **E1.1 (P0)** Email+password and Google OAuth signup; email verification via SendGrid; password reset.
- **E1.2 (P0)** Mandatory risk acknowledgement + ToS/Privacy consent (versioned, timestamped) before any trading feature.
- **E1.3 (P0)** Onboarding quiz ("experience? goal? market?") that routes to one of three start paths: Learn (Academy), Practice (paper), Connect (broker). Skippable.
- **E1.4 (P0)** 2FA (TOTP) available to all; **required** before the first live-broker connection.
- **E1.5 (P1)** Profile: display name, avatar, country (drives compliance flags §16), timezone (drives session filters).
- **E1.6 (P1)** Session management: active-device list, revoke, 30-day refresh tokens.
- **E1.7 (P2)** Phone/OTP login for MENA/India markets.

AC (E1.3): Given a new verified user, when they complete the quiz choosing "no experience," then they land in Academy lesson 1 with a paper account pre-created, and event `onboarding_path_selected` fires with path=learn.

E2 — Academy: Zero to Hero (Phase 0 v0 → GA full)

Goal: the acquisition hook and the churn-reducing habit loop.

- **E2.1 (P0)** Course structure: tracks → modules → lessons (video + text + images) → quizzes; progress tracked per user.
- **E2.2 (P0)** v0 content: 10-lesson "Foundations" free track (markets, candles, orders, risk 101, psychology, paper-trade walkthrough).
- **E2.3 (P0)** Quiz gating: pass ≥ 70% to mark lesson complete; completion awards points (§E12).
- **E2.4 (P0)** Admin CMS: create/edit/publish lessons, attach media, reorder, set free/paid flag per lesson — no deploys.
- **E2.5 (P1)** "Learn-then-do" deep links: lessons end with a CTA that opens the exact paper-trading screen pre-configured (e.g., "place your first limit order").
- **E2.6 (P1)** Certificates (shareable image) on track completion, with disclaimer footer.
- **E2.7 (P1, GA)** Full curriculum: Beginner (free+paid), Intermediate (risk, strategy, journaling), Automation (signals, webhooks, guards), Prop-firm track.
- **E2.8 (P2)** Cohort classes and live sessions; AR/HI localization.

AC (E2.5): Given Aisha completes "Order Types," when she clicks "Try it now," then a paper ticket opens pre-filled with a limit order on EURUSD demo and event `lesson_cta_practice` fires.

E3 — Paper Trading (Phase 0)

Goal: risk-free practice with real market prices; the activation engine.

- **E3.1 (P0)** Virtual account per user (\$10,000 default, resettable) with realistic execution against live/delayed prices (forex majors + top-20 crypto at v0; delayed feeds acceptable, latency disclosed).
- **E3.2 (P0)** Order types: market, limit, stop, stop-limit; SL/TP attach; partial close.
- **E3.3 (P0)** Full parity with the live trade ticket (§E6) so skills transfer 1:1.
- **E3.4 (P0)** PnL, equity curve, open positions, history — same dashboard components as live (§E18).
- **E3.5 (P1)** All signals, webhooks, and guards can target paper accounts (safe rehearsal for automation).
- **E3.6 (P1)** Monthly paper leaderboard (opt-in, display-name only) feeding tournaments (P2).
- **E3.7 (P2)** Configurable slippage/commission simulation profiles.

AC (E3.5): Given Marco points his TradingView webhook at his paper account, when an alert fires, then a simulated order executes and appears in paper history flagged

`simulated=true`), and no broker call occurs.

E4 — Broker & Exchange Connectivity (Phase 0: MT5 · Open Beta: CCXT · GA: cTrader)

Goal: connect in minutes; the moat is breadth (MT5 + cTrader + CEX) behind one abstraction.

- **E4.1 (P0) Connector abstraction layer** (`IBrokerConnector`): uniform interface for connect, account info, positions, orders, place/modify/cancel, history, streaming. All product features consume only this interface.
- **E4.2 (P0) MT5 via MetaAPI**: user supplies account number, password (investor password rejected for trading), server; provisioning, state polling, error surfacing (wrong password, server not found, broker restrictions).
- **E4.3 (P0) Connection lifecycle**: connecting → synchronizing → live → degraded → disconnected; visible status chip everywhere trading actions exist; auto-reconnect with backoff.
- **E4.4 (P0) Tier gating**: live connection attempts beyond plan limit are blocked server-side with upgrade CTA (E7.5).
- **E4.5 (P0) Credential security**: secrets encrypted with envelope encryption (KMS master key); never logged; never returned to client after save; delete = revoke at provider + purge.
- **E4.6 (P0, Open Beta) CEX via CCXT**: Binance + Bybit first; API key + secret with **trade + read scopes only — keys with withdrawal permission are rejected at validation**; IP-allowlist instructions surfaced.
- **E4.7 (P1, GA) cTrader Open API**: OAuth connect, account read + market/limit orders v1.
- **E4.8 (P1) Connection health page**: last sync, latency, error log, "test order" (paper route).
- **E4.9 (P2) Additional exchanges** (OKX, KuCoin, MEXC) config-driven via CCXT; additional MT5 capacity via self-hosted bridge (Phase 3 cost lever).

AC (E4.6): Given Chen submits a Binance key with withdrawal permission enabled, when validation runs, then the connection is rejected with the exact remediation message ("create a key with only Reading + Spot Trading"), and nothing is stored. AC (E4.2): Given valid MT5 credentials, when provisioning completes, then balance/equity/positions render within ≤ 3 minutes and event `broker_connected` fires with `provider=metaapi, ms_to_live`.

E5 — Account Mirroring & Sync (Phase 0)

Goal: HeroPips is the trusted mirror — every balance, position, order, and historical deal stored locally.

- **E5.1 (P0) Initial full sync on connect**: balance, equity, margin, open positions, working orders, and closed-deal history (up to 5 years where the provider allows).

- **E5.2 (P0)** Streaming updates (MetaAPI websocket / CCXT websockets) for order fills, position changes, balance changes; UI reflects within ≤ 3 s of provider event.
- **E5.3 (P0)** Reconciliation job every 5 min per live account: diff provider truth vs local store; discrepancies auto-corrected and logged as `(sync_anomaly)`.
- **E5.4 (P0)** Local persistence in PostgreSQL (accounts, positions, orders, executions/deals) with provider-native IDs preserved for idempotency and audit.
- **E5.5 (P1)** Export: CSV of history and positions per account.
- **E5.6 (P1)** Multi-account portfolio rollup (Pro/Elite): aggregate equity and PnL across all connections.
- **E5.7 (P2)** Retention: trade history retained ≥ 7 years (compliance-friendly default), user-deletable per GDPR flow (§16).

AC (E5.3): Given a position closed directly inside the MT5 terminal (outside HeroPips), when the next reconciliation runs, then local state matches the broker within 5 min and the change is attributed `(source=external)`.

E6 — Trade Ticket & Order Management (Phase 0)

Goal: one universal ticket for paper and every live venue, with full pre-submit control.

- **E6.1 (P0)** Universal order ticket: symbol search (per connected venue), side, order type (market/limit/stop/stop-limit), quantity or lot size, price(s), SL/TP, time-in-force, comment tag.
- **E6.2 (P0)** Pre-submit modification: user can change any parameter — type, quantity, price, SL/TP — before confirmation; ticket shows estimated margin, pip/tick value, and risk in account currency.
- **E6.3 (P0)** Position management: close (full/partial), reverse, modify SL/TP on open positions; cancel/modify working orders.
- **E6.4 (P0)** Every order action produces a local order record with lifecycle states (created $\rightarrow$ routed $\rightarrow$ accepted $\rightarrow$ filled/partial/rejected/cancelled) and provider IDs.
- **E6.5 (P0)** Human-readable rejection reasons mapped from provider errors (e.g., "Not enough margin", "Market closed", "Invalid stops level").
- **E6.6 (P1)** One-click **Panic Close**: close all positions and cancel all orders on an account (double-confirm; available on paper for Free).
- **E6.7 (P1)** Quantity helpers: fixed lots, % of equity, or risk-% (auto-computed from SL distance — powered by §E9 sizing).
- **E6.8 (P2)** Bracket/OCO templates; hotkeys.

AC (E6.2): Given Rahul opens a market order ticket with risk-% mode on, when he moves the SL, then the lot size recalculates live and the displayed risk stays at his configured % $\pm$ rounding to broker lot step.

E7 — Signal Engine: Predefined + Configurable + AI Tips (Closed Beta)

Goal: signals as configurable tools the user owns — never as advice.

- **E7.6 (P0) Predefined signal library** (admin-managed): 8-12 launch strategies (e.g., EMA cross trend, RSI mean-reversion, London breakout, BTC momentum), each with transparent rules, default risk settings, and historical stats labeled "hypothetical, not a recommendation."
- **E7.7 (P0) Signal instance = user's own configuration:** user selects a strategy, sets symbols, timeframe, session window, risk-% or fixed size, SL/TP logic, max concurrent trades, and target account (paper or live).
- **E7.8 (P0) Execution modes per instance:** (a) Notify-only, (b) One-tap confirm (push/email link, 15-min expiry), (c) Full-auto. Full-auto requires: explicit opt-in text, an active drawdown guard on the account, and re-confirmation every 90 days.
- **E7.9 (P0) Custom signal builder v1:** condition builder over indicators (EMA, SMA, RSI, MACD, ATR, price levels) with AND/OR groups, entry/exit rules; saved as versioned JSON config.
- **E7.10 (P0) Signal evaluation service:** evaluates enabled instances on candle close per timeframe; emits `signal_fired` events into the execution pipeline (§12) with dedupe key = (instance, symbol, bar time).
- **E7.11 (P0) AI trade tips (assistant, not advisor):** on ticket and on fired signals, generate contextual educational tips — SL/TP sanity vs ATR, risk-% vs account, session/news warnings, correlation warning across open positions. Every tip carries the disclaimer microcopy and links to a lesson.
- **E7.12 (P1) Signal instance dashboard:** status, last evaluation, fired history, per-instance PnL attribution, pause/resume.
- **E7.13 (P1) Delayed free preview:** Free tier sees predefined signals fire with 15-min delay and no execution (upgrade CTA).
- **E7.14 (P2) ML-ranked setups and AI-generated strategy suggestions** (kept in P2 pending compliance review — see Open Questions).

AC (E7.8): Given full-auto mode requested without a drawdown guard active, when the user tries to enable, then enabling is blocked with inline flow to create the guard first. AC (E7.10): Given the same bar fires twice due to service restart, when the second evaluation emits, then the execution pipeline drops it as a duplicate (idempotency key match) and logs `signal_duplicate_dropped`.

E8 — TradingView Webhook Automation (Closed Beta)

Goal: alert-to-order in seconds, exactly once, with the risk layer always in between.

- **E8.1 (P0) Per-user webhook endpoints:** unique URL + per-endpoint HMAC secret; user maps each endpoint to a target account (paper or live) and default order template.
- **E8.2 (P0) Payload schema v1** (documented, with copy-paste TradingView alert templates): action (buy/sell/close/close_all), symbol, qty|risk_pct, order_type, price?, sl?, tp?, comment?, idempotency_key?.
- **E8.3 (P0) Security:** HMAC signature verification (also accept secret-in-payload for TradingView's limitation, over TLS only), IP rate limits, replay protection (timestamp

+ nonce window 5 min), per-tier monthly alert budgets (E7 §7.1) with clear 429 behavior.

- **E8.4 (P0) Idempotency:** duplicate deliveries/retries with the same idempotency key (or same alert fingerprint within 10s) execute at most once.
- **E8.5 (P0) Symbol mapping:** per-endpoint map from TradingView tickers to broker symbols (e.g., `OANDA:EURUSD → EURUSD.a`), with auto-suggestions and validation.
- **E8.6 (P0)** Every webhook passes through the **risk engine** (§E9) before routing; guard rejections return 200 with `status=blocked_by_guard` and notify the user.
- **E8.7 (P1)** Webhook console: live log of received payloads, parse result, decision (executed/blocked/error), and broker response; test-fire tool with sample payloads.
- **E8.8 (P1)** Close-management verbs: `close_partial` (pct), `move_sl`, `flatten_symbol`.
- **E8.9 (P2)** Generic webhook support for other alert sources (indicators/EAs) with the same schema.

AC (E8.4): Given TradingView delivers the same alert three times within 8 seconds, when processing completes, then exactly one broker order exists and the console shows one executed + two deduplicated entries. AC (E8.3): Given an unsigned or badly signed payload, when it arrives, then it is rejected 401, nothing is queued, and repeated failures trip a temporary block on the source IP.

E9 — Risk Automation Suite "Trade Guard" (Closed Beta v1 → GA full)

Goal: the retention moat — enforce the rules that keep accounts alive. Guards apply account-wide and evaluate before AND after execution.

Pre-trade guards (block/adjust orders):

- **E9.1 (P0)** Risk-% position sizing (auto lot from SL distance; rounds to broker constraints).
- **E9.2 (P0)** Max risk per trade cap; max concurrent trades; max exposure per symbol/currency.
- **E9.3 (P0) Daily-loss circuit breaker:** when realized+floating loss for the day hits X% or \$X, block new entries until next trading day (user-set; hard-stop variant closes all).
- **E9.4 (P1)** Session filter (only trade user-defined windows, timezone-aware) and **news filter** (block entries $\pm N$ min around high-impact events from an economic calendar feed).
- **E9.5 (P1) Slippage cap** on market orders: submit as marketable limit at max deviation where the venue supports it; otherwise verify fill vs quote and alert on breach.
- **E9.6 (P0)** Guard evaluation is server-side, synchronous in the execution path, and fails closed (if guard service is unavailable, live orders are blocked, not passed).

Post-trade automations (manage open positions):

- **E9.7 (P1)** Break-even automation (move SL to entry +offset at R:R trigger); trailing stop (fixed pips or ATR); partial take-profit ladder.

- **E9.8 (P1) Stale-order auto-cancel:** cancel working orders older than N minutes/hours or beyond price drift threshold.
- **E9.9 (P0) Max-drawdown watchdog:** at X% equity drawdown, execute the user's chosen action (alert / block entries / flatten all).

Prop-firm presets (Elite):

- **E9.10 (P1, GA) Challenge guard templates** (generic "1-step/2-step challenge" presets: daily loss %, max loss %, min days) that combine E9.1-E9.9 and cannot be loosened while a challenge tag is active without a 24h cooldown. (Firm names used descriptively only — see Open Questions Q6.)

AC (E9.3): Given a daily-loss breaker at 3% and current day loss 3.1%, when any new entry arrives (manual, signal, or webhook), then it is rejected with

`(blocked_by_guard:daily_loss)`, the user is notified once (not per attempt), and closing/reducing positions remains allowed. AC (E9.6): Given the guard service is down, when a live webhook order arrives, then it is queued-blocked and the user is alerted "automation paused — safety first," and no unguarded live order reaches a broker.

E10 — Backtesting (GA: runtime · Phase 2: full engine)

Goal: let users validate configurations against history; monetizes Pro.

- **E10.1 (P0, GA) Runtime signal backtest (Pro+):** for any signal config, run over selected symbol/timeframe/date range (up to 5 years v1) using stored OHLCV; returns equity curve, win rate, profit factor, max DD, trade list; results labeled hypothetical, fees/slippage assumptions shown.
- **E10.2 (P0, GA) Data service:** TimescaleDB hypertables of M1+ aggregated OHLCV for launch symbol set (28 FX pairs, metals, top-30 crypto), 1-15 years depending on source; nightly ingestion jobs (Dukascopy for FX, Binance data for crypto; equities via licensed vendor when added).
- **E10.3 (P1, GA) Queueing + quotas:** backtests run async with per-tier daily quotas (Pro 20/day, Elite 100/day); progress and cancel.
- **E10.4 (P2) Full backtesting engine:** portfolio-level, tick-quality option, parameter sweep/optimization, walk-forward, export.
- **E10.5 (P2) Paper-forward testing:** auto-run a config on paper for N days and report live-vs-backtest drift.

AC (E10.1): Given identical config, range, and dataset version, when a backtest is re-run, then results are byte-identical (deterministic) and stamped with `(dataset_version)`.

E11 — Signal Lists, Marketplace & Share Cards (Beta: cards/follow-lite · Phase 2: marketplace)

Goal: virality (cards) now; network effects (marketplace) later.

- **E11.1 (P0, Open Beta) ROI share cards:** branded, auto-generated images for paper results, completed challenges, streaks, and (opt-in) live monthly results; every card includes the risk disclaimer and never shows dollar amounts by default (percent

only); one-tap share to X/IG/TikTok/Telegram/WhatsApp with UTM-tagged deep link.

- **E11.2 (P1, Open Beta) Curated signal lists (follow v1):** admin-curated lists of predefined-signal instances users can "follow" — following clones the config into the user's own account with their own risk settings (self-directed, not copy-trading of another user's account).
- **E11.3 (P2) Marketplace:** users publish signal configs with verified stats; followers subscribe; platform take-rate 20%; publisher payouts ride the affiliate rail and its KYC (§E14).
- **E11.4 (P2) Verified-performance badges** (only account-linked, unedited stats can display "verified").

AC (E11.1): Given Aisha finishes a +12% paper month, when she taps Share, then a card renders in < 2s with her display name, % return, "Paper account" label, disclaimer strip, and her referral link, and event `share_card_created` fires.

E12 — Gamification & Hero Points (Closed Beta)

Goal: habit and progress mechanics with zero money-transmission exposure.

- **E12.1 (P0)** Earn points for: lesson/quiz completion, daily streaks, first paper trade, first connection, first guarded trade, referral milestones, tournament placement (P2).
- **E12.2 (P0) Points are non-cash and non-withdrawable** (hard product rule): redeemable only for subscription time, feature unlock trials, academy modules, cosmetic profile badges, and merch; ToS states points have no monetary value and expire after 12 months of inactivity.
- **E12.3 (P0)** Points ledger: append-only transactions (earn/redeem/expire/adjust) with reasons; admin-adjustable with audit trail.
- **E12.4 (P1)** Levels ("Zero → Hero" ranks) computed from lifetime points; drives profile badges and share cards.
- **E12.5 (P1)** Anti-abuse: per-action caps, velocity limits, device/duplicate detection; fraudulent points reversible.
- **E12.6 (P2)** Seasonal quests and tournaments with non-cash prize pools.

AC (E12.2): Given any user or admin flow, when searching the product for a "withdraw points" capability, then none exists; redemption catalog contains only non-cash items.

E13 — Referral Program, all users (Closed Beta)

Goal: high-K viral loop that stays on the right side of pyramid/MLM law.

- **E13.1 (P0)** Every user gets a referral link/code; attribution: last-click, 30-day cookie + code-at-signup fallback; self-referral blocked.
- **E13.2 (P0) Reward model (default):** two-sided, subscription-triggered — referee gets 1 month free Starter credit at first payment; referrer gets 1 month subscription credit (or equivalent points). **Rewards trigger only on genuine paid subscriptions, never on signups alone.**

- **E13.3 (P0) Depth is admin-configurable but policy-capped:** engine supports N levels with per-level reward definitions; product config default = 2 levels (L2 at reduced value, credits only); **any config > 2 levels or any cash-type reward requires a `legal_signoff` flag set by an admin with the Compliance role** — UI surfaces the warning text from §16.
- **E13.4 (P0)** Referral rewards post to the same ledger as points/credits (§E12/E14) — no separate balances.
- **E13.5 (P1)** Referral dashboard: clicks, signups, conversions, earned credits; shareable assets (links into E11 cards).
- **E13.6 (P1)** Fraud controls: same-device/payment-instrument detection, refund clawback (reward reverses if referee refunds within 30 days).
- **E13.7 (P2)** Waitlist mechanics for launches: queue-jump by referrals.

AC (E13.3): Given an admin sets depth=5 with cash rewards, when saving without the Compliance role + signoff flag, then the save is rejected and the attempt is written to the audit log.

E14 — Affiliate / IB Engine (Open Beta v1 → GA)

Goal: per-deal configurable partner economics — the professional rail, separate from consumer referrals.

- **E14.1 (P0)** Affiliate accounts are applied-for and admin-approved; distinct portal at partner-facing subdomain; KYC required before first payout (identity + payout details).
- **E14.2 (P0) Deal templates + per-affiliate overrides:** revenue-share % (recurring, with duration), CPA per paid conversion, hybrid, and **IB rebate pass-through** (share of broker per-lot rebate attributed to the affiliate's referred accounts); tiered thresholds supported (e.g., \$4/lot to 100 lots, \$5.50 to 500, \$7 beyond). All parameters JSON-schema-validated, versioned, effective-dated.
- **E14.3 (P0) Sub-affiliate hierarchy (master-IB):** optional per-deal; configurable split between master and sub; max hierarchy depth per deal (default 2, hard cap 3); commissions computed from actual attributed revenue/volume only.
- **E14.4 (P0) Double-entry commission ledger:** every accrual, adjustment, clawback, and payout is a balanced journal entry; immutable; exportable; is the single source of truth for partner balances.
- **E14.5 (P0)** Attribution: dedicated links/coupon codes; conversion webhooks from billing; broker-volume ingestion for IB rebates (CSV import v1 → broker API/portal automation P1).
- **E14.6 (P1)** Payouts: monthly cycle, min threshold \$50, manual approval queue, rails = bank/Wise/USDT (ops-executed v1; automated P2); invoices auto-generated; withholding/geo blocks configurable.
- **E14.7 (P1)** Partner dashboard: real-time clicks/signups/conversions/volume/accruals; marketing asset library.
- **E14.8 (P2)** Self-serve tier for small affiliates (auto-approved, standard deal only).

AC (E14.4): Given any point in time, when summing ledger legs per partner, then debits equal credits and the partner "payable" balance equals accruals – clawbacks – payouts exactly; a mismatch blocks the payout queue and pages ops. AC (E14.2): Given a custom deal for Sana (25% rev-share 12 months + \$5/lot pass-through, 1 sub-level at 30/70 split), when configured and activated, then all subsequent attributed events accrue per those parameters and are visible in her dashboard within 15 min.

E15 — Billing, Plans & Payments (Phase 0 skeleton → Open Beta live)

- **E15.1 (P0)** Stripe subscriptions (monthly/annual), trials, proration, dunning, tax IDs; fallback regional PSP evaluation (Open Question Q3).
- **E15.2 (P0)** Plan catalog driven by admin config (§E7.2); coupon/promo support; LTD sold as one-time SKU that provisions the Pro-Lifetime entitlement and decrements a global seat counter (hard stop at cap).
- **E15.3 (P0)** Entitlement sync: payment events update entitlements within 60s; failed payment → grace 7 days → downgrade per E7.3.
- **E15.4 (P1)** In-app upgrade/downgrade with immediate effect and clear proration display; billing history + invoices.
- **E15.5 (P1)** Refund policy engine: 14-day money-back on first subscription; auto-clawback of referral/affiliate accruals on refund.
- **E15.6 (P2)** Crypto payments (USDT) via processor; regional pricing.

E16 — Admin Console & Tenant Configuration (Phase 0 → continuous)

Goal: everything a human tunes lives in admin config, not code — this is also the white-label foundation.

- **E16.1 (P0)** RBAC roles: SuperAdmin, Ops, Content, Support, Compliance, Finance; every admin action audit-logged (who/what/before/after).
- **E16.2 (P0) Tenant config object (JSON, versioned, DB-backed):** branding (name, logo, colors, domain), feature flags, plan catalog, signal library enablement, referral config, affiliate deal defaults, compliance flags per country, email templates. v1 has exactly one tenant (HeroPips) but ALL reads go through tenant config — no hard-coded brand strings anywhere.
- **E16.3 (P0)** User management: search, view (entitlements, connections, guards, ledgers), suspend, impersonate-read-only (audited), GDPR export/delete.
- **E16.4 (P0)** Signal library management: create/edit predefined strategies, publish/retire, set default params and tier availability.
- **E16.5 (P1)** Ops dashboards: connection fleet health, webhook error rates, execution latency, MetaAPI spend per account, guard-block stats.
- **E16.6 (P1)** LTD seat console; promo management; announcement banners.
- **E16.7 (P2)** Tenant provisioning UI (create tenant, assign domain, clone config) — activates in Phase 3.

AC (E16.2): Given the tenant logo/color changed in admin, when any page reloads, then new branding renders with zero deploy and the change appears in the audit log.

E17 — Notifications & Communications (Closed Beta)

- **E17.1 (P0)** Transactional email via SendGrid (templated, tenant-branded): verification, security, connection events, order executed/rejected, guard triggered, signal fired (per mode), billing.
- **E17.2 (P0)** In-app notification center with read state; per-category user preferences (execution, signals, guards, learning, marketing).
- **E17.3 (P1)** Telegram bot notifications (execution + guard alerts) — high-value for target geos; web push.
- **E17.4 (P1)** Notification batching/rate limits (e.g., guard-block storms collapse into one summary per 10 min).
- **E17.5 (P2)** Lifecycle marketing journeys (onboarding nudges, churn-save) with suppression rules.

E18 — Dashboards, Reporting & Journal (Closed Beta)

- **E18.1 (P0)** Home dashboard: per-account equity/balance/PnL (day/week/month), open positions, working orders, recent automation activity, guard status.
- **E18.2 (P0)** Trade history with filters (account, symbol, source: manual/signal/webhook/external) and per-trade detail (timeline of events, applied guards, tips shown).
- **E18.3 (P1)** Auto-journal: every closed trade gets an entry (source, setup tag, R multiple, notes field, screenshot attach); weekly review summary email.
- **E18.4 (P1)** Analytics: win rate, profit factor, expectancy, drawdown, performance by symbol/hour/strategy; paper vs live comparison.
- **E18.5 (P2)** Public profile page (opt-in) showing verified stats — feeds marketplace (E11).

E19 — White-Label Multi-Tenancy (Phase 3 — architectural requirements now)

- **E19.1 (P0-architecture)** `(tenant_id)` on every domain table from day 1; all queries tenant-scoped via middleware + Postgres RLS.
- **E19.2 (P0-architecture)** All branding/copy/plan/feature reads flow through tenant config (E16.2); theming via CSS variables.
- **E19.3 (P2)** Tenant onboarding: custom domain + SSL automation, per-tenant SendGrid sender, per-tenant Stripe (Connect) account, isolated affiliate program, tenant-admin role scoped to own tenant.
- **E19.4 (P2)** Tenant billing to HeroPips: setup fee + monthly + usage (connected accounts) + optional rev share; enforced by platform-level entitlements.
- **E19.5 (P2)** Data isolation guarantee documented + tested (no cross-tenant reads possible; per-tenant export).

9. Key User Flows (happy paths + critical errors)

Flow A — Zero to first paper trade (activation): Signup → verify → quiz (path=Learn) → Lesson 1-3 → CTA "Try it" → pre-filled paper ticket → place order → confetti + points +

`activated=true`). Error path: quiz skipped → land on dashboard with "Start here" checklist (3 items).

Flow B — Connect MT5 → first live guarded order: Upgrade (or trial) → 2FA enforced → enter MT5 login/server → provisioning status screen (≤3 min) → sync complete → prompted to create Daily-Loss Guard (default 3%) → open ticket in risk-% mode → order routed → fill notification (in-app + email). Error paths: bad credentials (inline provider message + retry), broker unreachable (status=degraded, retry with backoff, support link).

Flow C — TradingView alert → execution: Create endpoint → copy URL + secret + alert template → map symbols → send test payload → console shows parsed+simulated result → switch endpoint to live account → real alert → HMAC verify → idempotency check → guard evaluation → route → fill → notify. Error paths: bad signature (401, console entry), budget exceeded (429 + email once/day), guard block (200 blocked_by_guard + notification).

Flow D — Enable predefined signal in confirm-mode: Browse library → open strategy detail (rules + hypothetical stats + disclaimer) → configure instance (symbols, risk-%, session, target=paper first) → enable → signal fires → one-tap confirm link → executes → after 2 weeks nudge: "ready for live? requirements: guard active + full-auto consent."

Flow E — Affiliate deal to payout: Sana applies → admin approves + configures custom deal (E14.2) → she shares links → attributed signups/conversions/volume accrue to ledger → month-end: ops reviews payout queue → approve → payout executed → invoice emailed. Error path: KYC incomplete → payout held with clear checklist.

Flow F — Paywall & upgrade: Free user taps "Connect broker" → credential form disabled behind plan sheet (comparison table) → selects Starter → Stripe checkout → entitlement live ≤60s → returns to exact connect step with state preserved.

10. Non-Functional Requirements

Area	Requirement
Latency	Webhook receipt → broker submit: $p50 \leq 1.5s$, $p95 \leq 3s$ (excluding broker/venue time). Signal bar-close evaluation → submit: $p95 \leq 5s$. UI reflects provider events $\leq 3s$.
Availability	Beta 99.5%; GA 99.9% for execution path (webhooks, guards, routing). Status page public from Open Beta.
Fail-safety	Execution path fails closed for live accounts (no unguarded orders); paper may fail open. Every external call has timeout, retry with jitter, and circuit breaker.
Scalability	v1 targets: 50k registered users, 3k connected accounts, 200k webhook alerts/mo, 1k concurrent signal instances — on the modular monolith (§12). 10× headroom via horizontal API workers + queue partitioning before any re-architecture.
Security	OWASP ASVS L2; envelope encryption (AES-256-GCM, cloud KMS) for all broker/exchange secrets; secrets never in logs; TLS 1.2+; 2FA before live connect; RBAC + full audit log on order actions, admin actions, config changes; rate limiting on auth, webhooks, and API; annual pen test before GA.
Privacy	GDPR-style rights (export, delete) regardless of geo; PII minimization; DPA with subprocessors (MetaAPI, Stripe, SendGrid, hosting, analytics).
Compliance-by-design	Country flag system can disable features per jurisdiction (e.g., signals real-time view, marketplace) driven by tenant config (§16).
Localization	EN at launch; copy externalized to i18n files day 1; AR (RTL-ready UI) + HI in Phase 2.
Compatibility	Responsive web (mobile-first breakpoints); Chrome/Safari/Edge/Firefox last 2 versions; no native apps in v1.
Data durability	RPO ≤ 15 min (WAL archiving + nightly snapshots); RTO $\leq 4h$; TimescaleDB market data reproducible from source archives.
Observability	Structured JSON logs w/ correlation IDs end-to-end (webhook→order), metrics (Prometheus/Grafana), Sentry errors, per-account MetaAPI cost telemetry.

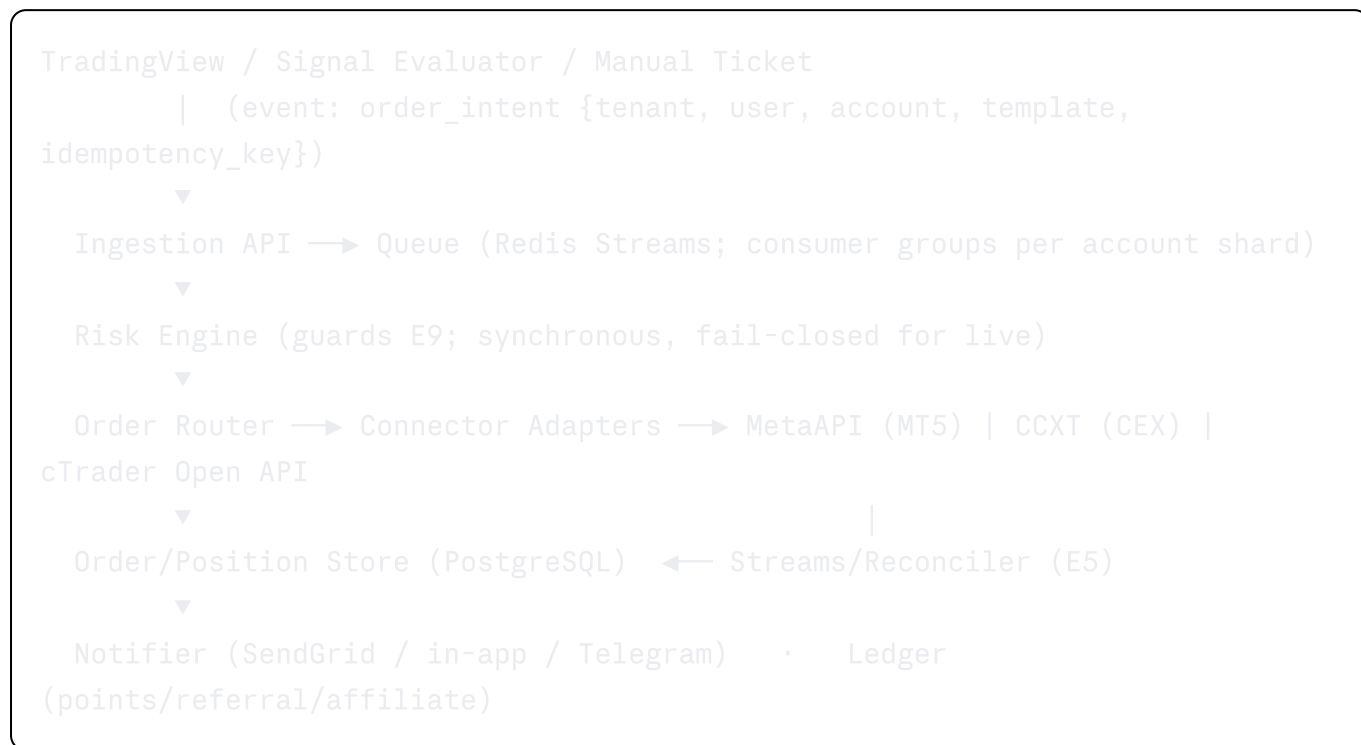
11. System Architecture (build spec summary)

Shape: modular monolith (NestJS/TypeScript) + worker processes, not microservices, until scale demands otherwise. One deployable API, clearly bounded internal modules mirroring the epics: identity, academy, paper, connectors, mirroring, execution, signals, webhooks,

guards, backtest, billing, ledger (points/referral/affiliate share one ledger core), admin/tenant, notifications, analytics.

Frontend: Next.js (App Router) on Vercel; design tokens from tenant config (CSS variables); TradingView Lightweight Charts for charting.

Core services & data flow (execution pipeline):



Queueing decision: Redis Streams at MVP (already needed for cache/rate-limits; lowest ops burden). Graduation trigger to Redpanda/Kafka: sustained > 200 msg/s, multi-consumer fan-out needs, or multi-region. The `order_intent` event schema is transport-agnostic so migration is a swap, not a rewrite.

Market data & backtest: TimescaleDB (Docker) with hypertables `ohlcv(symbol, tf, ts, o, h, l, c, v)` + continuous aggregates; ingestion workers (Dukascopy FX, Binance dumps); dataset versioning table; backtest workers consume read-only replicas.

Environments & deploy: Docker everything; docker-compose on Hetzner (2 app nodes + Postgres + Timescale + Redis) for beta; GitHub Actions CI/CD (lint, tests, migration gate, image build, deploy); IaC via simple Terraform; secrets in cloud KMS/SSM. Est. infra at beta: low hundreds USD/mo excluding MetaAPI per-account fees (tracked per \$10 observability). Kubernetes only when node count or team size forces it.

Key integrations: MetaAPI (MT5 provisioning/trading/streaming — cost is per connected account; gate per \$7), CCXT (Binance, Bybit v1), cTrader Open API (OAuth), Stripe, SendGrid, economic-calendar feed (news filter), PostHog (product analytics), Sentry, Cloudflare (WAF/rate-limit edge for webhook endpoints).

12. Data Model — core entities (abridged)

`tenants` · `users` (country, tz, flags) · `consents` (doc version, ts) · `plans` / `subscriptions` / `entitlements` · `broker_connections` (provider, status,

encrypted_secret_ref) · (accounts) (live/paper) · (positions) · (orders) (state machine, provider_ids, source: manual|signal|webhook|external, idempotency_key) · (executions) · (sync_anomalies) · (signals) (library) · (signal_instances) (user config JSON, mode) · (webhook_endpoints) (secret_ref, mapping) · (webhook_events) (payload, decision) · (guards (type, params, scope) · (guard_events) · (backtests) (config, dataset_version, results) · (courses/modules/lessons/progress/quiz_attempts) · (points_tx) · (referral_edges (referrer, referee, level) · (referral_rewards) · (affiliates) / (affiliate_deals) (versioned JSON) / (attributions) / (ledger_entries) (double-entry: account, direction, amount, currency, ref) / (payouts) · (share_cards) · (notifications) / (preferences) · (audit_log (actor, action, before/after) · Timescale: (ohlcv), (calendar_events).

Conventions: every domain table carries (tenant_id); money as integer minor units + currency; all state changes event-sourced to an outbox table consumed by the notifier/analytics.

13. Analytics & Instrumentation (PostHog event taxonomy v1)

(signup_completed) · (onboarding_path_selected{path}) · (lesson_completed{lesson_id}) · (paper_account_created) · (first_paper_order) · (paywall_viewed{trigger}) · (checkout_completed{plan}) · (broker_connect_started/succeeded/failed{provider, error}) · (first_live_order) · (guard_created{type}) · (guard_blocked{type}) · (signal_instance_enabled{strategy, mode}) · (signal_fired/executed/blocked) · (webhook_created) · (webhook_received/executed/deduped/blocked{reason}) · (backtest_run{duration, range}) · (share_card_created/clicked) · (referral_link_shared) · (referral_converted{level}) · (affiliate_conversion) · (churn_survey_submitted{reason}).

North-star: **Weekly Guarded Live Traders** (users with ≥ 1 live order passing ≥ 1 guard in the week). Funnels: signup→activation (24h), free→paid, connect→first-order, webhook-created→first-execution.

14. Legal, Compliance & Policy Requirements (product-enforced)

- **C1 (P0)** Every signal, tip, backtest result, share card, certificate, and marketing surface renders the standard risk disclaimer component; copy owned by counsel; version-tracked.
- **C2 (P0)** Positioning guardrails in-product: no UI copy may state or imply guaranteed profit; performance always labeled hypothetical or historical; a copy-lint checklist gates releases.
- **C3 (P0)** Full-auto execution requires explicit informed consent (dedicated screen, versioned text, re-consent every 90 days) + active drawdown guard (E7.8).
- **C4 (P0)** No custody: no user fiat/crypto balances on HeroPips; points are non-cash (E12.2); partner cash moves only via the KYC'd affiliate rail (E14).

- **C5 (P0)** Referral policy defaults per E13.3; exceeding depth 2 or adding cash rewards requires Compliance-role signoff recorded in audit log.
- **C6 (P0)** Country compliance flags: per-country toggles (e.g., disable real-time signal display, disable marketplace, block signup) managed in tenant config; US visitors blocked from signals/auto-trade features at launch (conservative default pending counsel).
- **C7 (P0)** IB rebate disclosure: any broker-link flow states that HeroPips may receive volume-based compensation from the broker.
- **C8 (P1)** ToS/Privacy acceptance re-prompt on material change; data-processing register maintained; subprocessor list published.
- **C9 (P1)** Marketing claims review queue for affiliate-submitted creatives.
- **Counsel to engage before Open Beta:** UAE structure (free zone now; DIFC/ADGM + ITL evaluation), robo-advice boundary review of E7.11/E7.14, referral program review, LTD consumer terms.

15. Rollout Plan, Milestones & Exit Criteria

Milestone	Date	Exit criteria (go/no-go)
M0 Foundation complete	Sep 30, 2026	MT5 connect + mirror + manual ticket work E2E on 3 real broker accounts; Academy v0 live; paper v0 live; audit log + entitlements enforced
M1 Closed Beta start	Oct 1, 2026	500 invites; crash-free sessions $\geq 99\%$; webhook pipeline passes idempotency + guard chaos tests
M2 Closed Beta review	Oct 31, 2026	Activation $\geq 30\%$; $\geq 60\%$ of connected users created a guard; NPS ≥ 30 ; execution p95 $\leq 3s$
M3 Open Beta + monetization	Nov 10, 2026	Stripe live; LTD live w/ seat counter; CCXT (Binance/Bybit) live; affiliate portal v1; status page up
M4 Beta ends	Dec 31, 2026	$\geq 8k$ registered; ≥ 250 paying (subs+LTD); churn signal read; sev-1 incidents = 0 in final 2 weeks
M5 GA v1.0	Jan 15, 2027	Runtime backtest (Pro) live; cTrader v1; pen test passed; counsel signoff on C1-C7; pricing finalized from beta data
M6 GA+90 review	Apr 2027	$\geq 3\%$ free $\rightarrow$ paid; MRR trajectory to \$40k by mid-2027; guard-user churn delta measured; decide Phase 2 order

Beta cohort plan: 40% beginners (P1), 30% MT5 traders (P2), 20% TradingView (P3), 10% crypto (P4) — recruited via Dubai/India/SEA communities and 5 seeded affiliate partners.

16. Product Risks & Mitigations (top 8)

Risk	Mitigation in this PRD
MetaAPI per-account cost destroys margin	Free tier = paper only (§7); per-account cost telemetry (§10); idle-connection auto-sleep (P1); self-hosted bridge in Phase 3
Unguarded automation harms a user → trust/regulatory blowup	Fail-closed risk engine (E9.6); full-auto consent + mandatory guard (E7.8, C3); complete audit trail (E6.4)
Referral program read as MLM	Depth/policy caps + Compliance gate (E13.3, C5); non-cash defaults
Bonus wallet = money transmission	Points non-cash by construction (E12.2, C4)
Duplicate/replayed webhooks double-fire orders	HMAC + replay window + idempotency keys + dedupe tests (E8.3-E8.4)
Losing traders churn regardless of product	Academy habit loop (E2), guards as dependency (E9), journal/insights (E18), annual plans (E15)
Exchange/broker API policy shifts	Connector abstraction (E4.1); ≥ 2 venues per asset class by GA; no single-provider feature coupling
Scope creep kills the December date	Non-goals (§4) are contractual; any addition requires equal removal (change-control note in §17)

17. Open Questions

#	Question	Owner	Blocking?
Q1	MetaAPI enterprise/volume pricing — get written quote incl. per-account fees at 500/2k/10k accounts	Founder/Eng	Blocks pricing final (M5), not build
Q2	Sequencing check: is cTrader demand real enough for GA, or swap with more CCXT venues? (Beta survey)	Product	Non-blocking; decide by M4
Q3	Stripe availability/limits for the UAE entity; fallback PSP (Checkout.com/Tap) and USDT payout rail for affiliates	Finance/Legal	Blocks M3
Q4	Counsel: does E7.11 (AI tips) or E7.14 (AI-suggested strategies) cross the robo-advice line in target jurisdictions? Exact disclaimer language	Legal	E7.11 blocks Closed Beta copy; E7.14 stays P2 until cleared
Q5	Country block list at launch (proposal: block US from signals/auto-trade; confirm CA/JP/IN treatment)	Legal	Blocks M1
Q6	Prop-firm preset naming: generic labels vs referencing firm names (trademark risk)	Legal	Non-blocking (Elite ships at GA)
Q7	Economic-calendar data source licensing for the news filter	Eng	Blocks E9.4 (P1)
Q8	LTD final price point and seat count (\$499 vs \$799; 300 vs 500) — decide from closed-beta willingness-to-pay survey	Founder	Blocks M3
Q9	Paper-trading price feed source and whether 15-min delay is acceptable for beginners v1	Eng/Product	Blocks M0 detail, default = delayed OK with label
Q10	Brand/trademark clearance for "HeroPips" in UAE, India, EU	Legal	Non-blocking; start now

18. Appendix

A. TradingView webhook payload v1 (example):

```
json
```

```
{
  "key": "whk_live_7f3a...",
  "idempotency_key": "tv-{{time}}-{{ticker}}-buy",
  "action": "buy",
  "symbol": "OANDA:EURUSD",
  "order_type": "market",
  "risk_pct": 1.0,
  "sl": "{{plot_0}}",
  "tp": "{{plot_1}}",
  "comment": "ema-cross-h1"
}
```

B. Signal instance config (stored JSON, abridged):

```
json
{
  "strategy_id": "ema_cross_v1",
  "version": 3,
  "symbols": ["EURUSD", "GBPUSD"],
  "timeframe": "H1",
  "session": {"tz": "Asia/Dubai", "windows": [["07:00", "19:00"]]},
  "risk": {"mode": "risk_pct", "value": 1.0, "max_concurrent": 3},
  "exits": {"sl": {"type": "atr", "mult": 1.5}, "tp": {"type": "rr", "value": 2},
  "mode": "confirm",
  "target_account": "acc_paper_01"
}
```

C. Glossary: IB = Introducing Broker (broker pays volume-based rebates for referred accounts) · Guard = server-side risk rule evaluated in the execution path · Instance = a user's configured copy of a signal strategy · LTD = lifetime deal · Fail-closed = on component failure, block live orders rather than pass them · Entitlement = server-side capability grant derived from plan · Tenant = a branded platform instance (HeroPips itself is tenant #1).

D. Change control: post-approval, scope changes require: written proposal → impact on M-dates → one-in-one-out trade → founder + eng-lead signoff → PRD version bump.

— End of PRD v1.0 —